

Chia-Heng Liu

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EDUCATION

Cornell University — Masters, Electrical and Computer Engineering 2025 – 2026

Research: Mixed-Precision AI Accelerators for LLM Inference

National Tsing Hua University — Bachelors, Materials Science and Engineering 2020 – 2024

Relevant Coursework: Complex Digital ASIC Design (Tapeout), Computer Architecture, Multicore Architecture, Memory-Centric Computing, AI Hardware, Systems for Large-Scale ML, Data Center Architecture, Computer Networks

Honors & Awards: Phi Kappa Phi Honor Society (Top 10% of graduate students, Cornell University)

EXPERIENCE

ASIC RTL Intern, Etched Summer - Fall 2026

- Owned RTL design and implementation of a destination-based router microarchitecture for chip-to-chip MoE token routing in a scale-out AI network, featuring dynamic expert-buffer allocation and continuous token dispatch to reduce TTFT and enable scalable expert dispatch across distributed accelerator nodes.
- Designed PnR-friendly reduction and broadcast interconnects for a wide I/O die chip-to-chip communication fabric, reducing cross-sectional wiring and enabling physically scalable implementation.
- Designed a debug packet generator/checker to exercise end-to-end datapaths across chip-to-chip links, die-to-die links, and I/O-die crossbars, enabling deployment-time validation and long-duration soak testing.

Research Assistant, NYCU CERES Lab 2024–2025

- Implemented a stochastic-computing MAC architecture for neural network accelerators achieving 1049 μm^2 area and 0.645 mW power in TSMC 40 nm.
- Co-op Project with ITRI: developed a NoC-based AI accelerator simulator to evaluate low-power data movement and reuse strategies.

SELECTED PROJECTS

ASIC Projects: Mixed-Precision MLP Accelerator (Academic Tapeout)

SystemVerilog, Synopsys (VCS, Design Compiler, PrimeTime), Cadence Innovus, Mentor Calibre 2025 – Present

- Architected and taped out a 32-lane precision-scalable nonlinear accelerator in TSMC 180 nm.
- Integrated a 32×32 INT8 GEMM array with vector and nonlinear compute units.
- Implemented activation caching, streaming dataflow, and self-checking verification infrastructure, validating execution on quantized CNN, MLP, and LLM inference workloads.
- Achieved $21 \times$ speedup and $8.6 \times$ energy reduction on GELU workloads versus a similarly sized RISC-V processor, with $<1\%$ accuracy degradation on SST-2 and $\sim 2\%$ on SQuAD in LLM inference simulations.

CPU Project: 4-Core Pipelined RISC-V CPU with NoC and Write-Back Cache

SystemVerilog, Verilator, PyMTL3 Sep. 2025 – Nov. 2025

- Implemented a 4-core RISC-V processor with router-based NoC, hazard detection, forwarding logic, and a 2-way write-back cache.
- Verified multicore execution using PyMTL3 and Verilator, achieving $2.5 \times$ speedup on parallel workloads.

PUBLICATIONS

Chia-Heng Liu et al., Energy-Efficient and Accurate Stochastic Computing-based Multiply-Accumulate Architecture for Neural Network Accelerator Design (First-Author) *IEEE MCSoc 2025*

Chia-Heng Liu et al., FlexNLU: A Window-Adaptive Stochastic Nonlinear Engine for Mixed-Precision Edge Transformers (First-Author) *Manuscript in preparation*

TECHNICAL SKILLS

- **Hardware:** SystemVerilog/Verilog, RISC-V, NoC, AI Accelerators, Parallel Compute Datapaths, Spice
- **EDA Tools:** Synopsys (VCS, Verdi, Design Compiler, PrimeTime), Cadence (Innovus, Virtuoso), Mentor Calibre
- **Programming:** C++, Python, PyTorch, Linux, TCL